

Teacher Guide — primary / module-11-multiplication-near-base

IKS Curriculum

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Overview

Module 11 — The Multiply-Near-Ten Trick (Towards Nikhilam)

Age band: Primary · **Grades:** 3–5 · **Duration:** 10 sessions × 30 min

What children will learn

1. The deficits from 10: $1 \rightarrow 9, 2 \rightarrow 8, 3 \rightarrow 7, 4 \rightarrow 6, 5 \rightarrow 5, 6 \rightarrow 4, 7 \rightarrow 3, 8 \rightarrow 2, 9 \rightarrow 1$. Instantly.
2. A finger trick (the “hand multiplication” trick) for multiplying any two digits from 6 to 9.
3. The idea that you can multiply two “near 10” numbers by combining the deficits.
4. A peek at the full Nikhilam method (*Nikhilam*) as a “trick the older kids will learn.”
5. A drawing project: design a poster or comic that teaches the finger trick to a younger child.

How this module sounds

- Lots of dot patterns. Finger play. Chanting.
- No long worksheets. Children draw, count, talk, chant.
- Sanskrit term *Nikhilam* introduced once on Day 6 as “the older-grade name for this trick.”

NEP 2020 alignment

- §5.1 — Foundational numeracy.
- §4.6 — Indian Knowledge Systems.
- §4.23 — Experiential learning (hand games, drawing, poster-making).

Prerequisites

- Comfortable with multiplication facts up to 5×5 .
- Can count fingers up and down with both hands.
- Can write single digits clearly.

Materials

- Class wall-poster: “Deficits from 10” ($1 \leftrightarrow 9$, $2 \leftrightarrow 8$, $3 \leftrightarrow 7$, $4 \leftrightarrow 6$, $5 \leftrightarrow 5$).
- A4 drawing paper, crayons, coloured pencils.
- Counters (beads or buttons) — 10 per child.
- Dice (one per pair).
- Finger-trick demo card (teacher prepared, see Day 3).

Day-by-day

Day	Title	What we do
1	Make the Twin	Card game: find the “twin” that makes 10 with each card
2	Twins Chant	Memorise the 9 deficit pairs
3	The Finger Trick	Hand-multiplication for $6-9 \times 6-9$
4	Why the Trick Works (drawing)	Dot patterns showing $7 \times 8 = 56$
5	Pairs Race	Speed game with the finger trick
6	Meet Nikhilam	Brief demo by teacher of 97×96
7	Design a Game	Children design a Make-the-Twin game
8	Project Day 1	Start a poster or comic of the finger trick
9	Project Day 2	Finish + decorate
10	Showcase	Stick posters on wall; each child explains theirs in 1 minute

Project

Children design a **poster or 4-panel comic** that teaches the finger-multiplication trick (for $6-9 \times 6-9$) to a younger child. They share it on Day 10.

See `assessments/project-brief.md`.

Sources

The finger-multiplication trick is a folk method taught widely in Indian schools and globally. The Sanskrit name *Nikhilam* and the full multiplication procedure come from Bharati Krishna Tirthaji’s *Vedic Mathematics* (1965). For older grades’ deeper treatment, see the Middle-band version of this module.

10-Day Lesson Plan

Module 11 (Primary) — 10-Day Lesson Plan

At a glance (30-min sessions)

Day	Title	Activity	Output
1	Make the Twin Game	Pair card game	First-day fun
2	Twins Chant	Memorise the 9 pairs	Chant + recall
3	The Finger Trick	Hand multiplication 6×6 to 9×9	First mental answer
4	Why the Trick Works	Dot-pattern drawing of $7 \times 8 = 56$	Drawing pasted
5	Pairs Race	Speed game with finger trick	Personal best
6	Meet Nikhilam	Teacher demo of 97×96	Awe
7	Design a Game	Children design own poster / game	Sketch
8	Project Day 1	Build the poster	Poster halfway
9	Project Day 2	Finish + colour	Poster ready
10	Showcase	Wall display + 1-min share	Stickers

Daily structure (30 min)

Time	What
5 min	Settle + the “Twins” chant
7 min	Story / new game introduction
13 min	Game / activity
5 min	Sharing / reset

The daily chant

“1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!”

(Same as Module 5’s “Make 10” chant — reuse it.)

Sing it / chorus it with claps. By Day 3 it should be effortless.

Pacing notes

- Games > worksheets at this band. If a worksheet feels slow, drop it and add another game round.
- Day 6 (Nikhilam demo) is a “wow” moment — keep it short and mysterious.

- Days 8–10 build to the poster showcase. Confirm pair groupings by Day 6.

What children leave with

- Instant recall of all 9 deficit pairs.
- A finger trick they can use to multiply any two digits from 6 to 9.
- Curiosity about the older-grade trick (Nikhilam).
- One poster or comic they made and shared with classmates.

Teacher Notes

Teacher Notes — Module 11 Multiply-Near-Ten (Primary)

Posture

This module gives Primary-band children a *single concrete win*: by Day 5 they can multiply any two digits from 6 to 9 using a finger trick. That’s it. The full Nikhilam algorithm is reserved for Middle band. The finger trick IS Nikhilam at base 10 — but we don’t make that connection explicit until Day 6, and even then it stays as “an older-grade name.”

Common things to handle

1. **A child can’t yet recall all the deficit pairs.** That’s fine — the module is the practice. Don’t shame; bring them into more games.
2. **A child gets bored (“I already know my times tables!”).** Give them the “speed challenge”: all 16 problems (6×6 through 9×9) in under 30 seconds using the finger trick.
3. **A child asks “why does the finger trick work?”** Show the dot-pattern drawing on Day 4. If they want more, give them the *algebra hint*: “your fingers UP $\times$ fingers UP = the units. That’s the deficit $\times$ deficit from older grades. You’ll see it again.”
4. **A child says “ 9×9 doesn’t work on my fingers!”** It does — both hands have 1 finger down (deficit 1 for each). Fingers down = 2 $\rightarrow$ that’s the tens (20). Fingers up = 4 left, 4 right $\rightarrow 4 \times 4 = 16 \rightarrow$ units. Total $20 + 16 = \dots$ wait, the kids will need help. Show them the standard form: 9×9 : deficits 1, 1; left $9 - 1 = 8$; right $1 \times 1 = 1 \rightarrow$ **81**. The finger trick variation needs adapting for the smallest deficits.

The finger trick (canonical form)

For $a \times b$ where both a and b are in 6–9:

1. Hold up both hands, palms toward you.
2. On the **left** hand, fold down $(a - 5)$ fingers. Fingers DOWN on left = $a - 5$. Fingers UP on left = $10 - a$.
3. On the **right** hand, fold down $(b - 5)$ fingers. Fingers DOWN on right = $b - 5$. Fingers UP on right = $10 - b$.

4. **Sum of fingers DOWN** on both hands = the TENS digit ($\times 10$).
5. **Product of fingers UP** on both hands = the UNITS digit.
6. **Tens + Units = answer.**

Example 7×8 : - Left: fold down 2 fingers ($7 - 5 = 2$). 2 down, 3 up. - Right: fold down 3 fingers ($8 - 5 = 3$). 3 down, 2 up. - Down: $2 + 3 = 5$. $\rightarrow$ **50**. - Up: $3 \times 2 = 6$. $\rightarrow$ **6**. - Answer: $50 + 6 = 56$. ✓

Example 6×7 : - Left: 1 down, 4 up. Right: 2 down, 3 up. - Down: $1 + 2 = 3$. $\rightarrow$ 30. - Up: $4 \times 3 = 12$. $\rightarrow$ 12. - $30 + 12 = 42$. ✓

Example 9×9 : - Left: 4 down, 1 up. Right: 4 down, 1 up. - Down: $4 + 4 = 8$. $\rightarrow$ 80. - Up: $1 \times 1 = 1$. $\rightarrow$ 1. - $80 + 1 = 81$. ✓

The connection to Nikhilam (don't tell kids until Day 6)

- Fingers DOWN on left = $a - 5$. Fingers UP on left = $10 - a$ = deficit of a from 10.
- Sum of fingers down = $(a - 5) + (b - 5) = a + b - 10$. Times 10 = $10a + 10b - 100$.
- Product of fingers up = $(10 - a)(10 - b)$ = deficit $\times$ deficit.
- Sum: $10a + 10b - 100 + (10 - a)(10 - b) = 10a + 10b - 100 + 100 - 10a - 10b + ab = ab$. ✓

So the finger trick **IS** Nikhilam — just with a different mnemonic that maps deficits to fingers-up. Children don't need this proof. The Middle module covers it.

Safety

- No physical risk in this module.

Differentiation

- **Tier up:** Grade 5 children may be ready for 2-digit $\times$ 1-digit problems using the deficit idea (e.g. 8×12). Or extend the finger trick to base 5 (multiplying 3–4 by 3–4 using a 1-hand variant).
- **Tier down:** Grade 3 children still building number sense need EXTRA game time on Days 1–2. Skip the finger trick on Day 3 and do another twins game; introduce the finger trick on Day 4 with the dot drawing as foundation.

Project oversight

- Days 7–10: children make a poster or 4-panel comic.
- Materials: A4 sheet, crayons. Don't expect families to provide elaborate supplies.
- If a child can't draw, they can write the steps in big letters with arrows.

Honest history — for parent or older-sibling questions

If a parent or older sibling asks: “*Is this real Vedic mathematics?*” — your answer is: > “The finger trick is folk math taught all over the world; the Sanskrit name *Nikhilam* comes from a

1965 book by Tirthaji. The math is real and useful. We teach the technique here; the history we engage with in the older grades.”

If you only have 6 days

- Day 1: Make the Twin Game.
- Day 2: Twins chant + introduce finger trick.
- Day 3: Finger trick + dot pattern.
- Day 4: Meet Nikhilam (the wow demo).
- Day 5: Project day (poster).
- Day 6: Showcase.

Sources

Sources — Module 11 Multiply-Near-Ten (Primary)

For Primary band, the teacher-facing sources are documented here. Children do **not** see citations in their lesson materials at this band (per style-guide).

Where the finger trick comes from

The “hand multiplication” or “finger multiplication” trick for digits 6–9 is a folk method taught widely in schools across India and the world. It has no single named “first source” — it predates formal records. The algebra behind it is exactly the *Nikhilam* algebra $((10 - a)(10 - b) = 10(10 - a - b) + ab)$ — fingers-down = deficits, fingers-up = the deficit-product.

Where the Sanskrit name comes from

- **Tirthaji, Bharati Krishna.** *Vedic Mathematics: Sixteen Mathematical Formulae from the Vedas* (1965), Motilal Banarsidass. — The Sanskrit name *Nikhilam Navataścaramaṁ Daśataḥ* (“all from 9 and the last from 10”) was popularised by Tirthaji in this posthumously-published book.

For the Middle / Senior pipeline

The full Nikhilam multiplication procedure (2-digit × 2-digit and beyond) is in: - *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center, Sammamish WA) — in `/Users/sree/Projects/SutraGanita/content/`. - *Microsoft Word - sutras1.pdf* — the 16 Tirthaji sūtras.

These are teacher references at this band, not student-facing.

Honest history (for teachers, if asked)

Tirthaji claimed his sūtras came from an *Atharva-veda Pariśiṣṭa* (Vedic appendix text). No other scholar has produced this manuscript. Mainstream historians (Dani 1993, Plofker 2009) consider the system 20th-century synthesis rather than strictly ancient.

For Primary band the framing is simple: “*This trick has a Sanskrit name. The math is real. The history is older when you’re older.*”

Citation discipline at this band

- No PDFs cited in student-facing slides, workbooks, or day-files.
- The Sanskrit term *Nikhilam* appears ONCE (Day 6) as cultural anchor — meaning given inline.
- All scholarly framing kept in `teacher-notes.md` and this file.

Day 1 — Make the Twin Game

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children play a card game finding the “twin” — the partner that makes 10 with each card.

Materials

- A deck of number cards 1–9 (4 of each = 36 cards) per pair.
- Counters (optional).

The flow

Settle + introduce the chant (5 min)

Teach the chant — clap on each pair:

“1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!”

Say it 3 times. (Children from Module 5 will know this — re-use it.)

Story (5 min)

“*Today we play a game called Make the Twin. Every number from 1 to 9 has a twin that makes 10 with it. 1’s twin is 9. 2’s twin is 8. Today we find them.*”

Game (17 min)

How to play:

1. Shuffle the deck. Deal 5 cards to each player. Place 5 cards face-up in the middle.
2. On your turn:
 - Look at your hand and the middle.
 - Find a “twin pair” — your card + a middle card that add to 10.
 - Take both. Put them in your “pile.”
 - Draw a new card from the deck.

3. If you can't find a twin, place 1 card in the middle. End your turn.
4. When the deck is empty, count pairs. Most pairs wins!

Play 2 rounds.

Show + chant (3 min)

- “Who won a round?” — winner shares ONE twin pair they made.
- Class chants the pair: “*6 and 4 make 10!*”
- Close with the daily chant.

Homework

- At home, practise the chant with a family member. Win a round of Make the Twin if you can find a partner!

Differentiation

- **Tier up:** Add cards 0 and 10. Find twins for those too. ($0 + 10$. Or just play with 1–9.)
- **Tier down:** Use only cards 1–6 to keep the twins easy.

Teacher's quick notes

- Some children won't find twins and will give up. Pair them with a stronger player as a “team.”
- The game IS the lesson at this band. Don't add a worksheet.
- Note: children who completed Module 5 will be fast at twins. Push them with the “speed challenge” — name the twin in under 1 second.

Day 2 — Twins Chant

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children commit the 9 deficit-from-10 pairs to memory and can recall any twin in under 1 second.

Materials

- Class wall-poster: “Twins of 10” (pre-made; show all 9 pairs).
- Index cards with single digits 1–9 (one set per child).
- Stopwatch.

The flow

Settle + chant (5 min)

Sing the chant:

“1 and 9 make 10! 2 and 8 make 10! 3 and 7 make 10! 4 and 6 make 10! 5 and 5 make 10!”

Three times, with claps.

Now add the “deficit” version (just renaming):

“1’s twin is 9! 2’s twin is 8! 3’s twin is 7! 4’s twin is 6! 5’s twin is itself!”

Story (5 min)

- “*Today we make the twins so fast we don’t even think. We just KNOW.*”
- Hold up the wall poster. Walk through the 5 pairs. Notice: each pair appears twice in the “twin” naming — 1’s twin is 9 AND 9’s twin is 1.

Lightning round (10 min)

- Teacher holds up a digit. Class chorus the twin.
- Do 20 in 60 seconds.
- Repeat with random sequence.

Sequence example:

7 → 3
8 → 2
6 → 4
9 → 1
5 → 5
4 → 6
3 → 7
2 → 8
1 → 9
8 → 2
6 → 4
7 → 3
...

Card sort (7 min)

- Each child gets index cards 1–9.
- On “Go,” they arrange the cards into 5 pairs (with 5’s pair being itself or stand-alone).
- Fastest 3 children share their arrangement on the floor.

Show + chant (3 min)

- Final chant. Pull the index cards back into the deck.

Homework

- At home, have someone call out 10 digits in 30 seconds. You say the twin. Beat 8/10.

Differentiation

- **Tier up:** Speed of 30 twins in 30 seconds.
- **Tier down:** Slow chant. 5 twins in 30 seconds is fine for today.

Teacher's quick notes

- Pay attention to children who PAUSE before answering. They haven't built the reflex yet. Pull them into more drills tomorrow.
- The single most important fluency in this module. Don't move on if half the class is still pausing.

Day 3 — The Finger Trick

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children learn the hand-multiplication trick for any pair from 6 to 9.

Materials

- Both hands (everyone has them).
- Class wall-poster: a hand diagram showing fingers folded down for 7×8 .
- Worksheet: 16 problems (all combinations 6×6 through 9×9), with hand-diagram answer key.

The flow

Settle + chant (5 min)

The Twins chant + lightning round of 10 twins.

Story (8 min) — “The Magic Hand Trick”

- “Today I’ll teach you a magic trick. You can multiply 7×8 just by counting fingers. Watch.”
- **Demo: 7×8 .**
 - “Hold up both hands. We’re going to fold down some fingers.”
 - “For 7: fold down $(7 - 5) = 2$ fingers on the LEFT hand. So 2 down, 3 up on the left.”
 - “For 8: fold down $(8 - 5) = 3$ fingers on the RIGHT hand. So 3 down, 2 up on the right.”
 - “Count the fingers DOWN on both hands together: $2 + 3 = 5$. That’s the TENS. → **50.**”
 - “Count the fingers UP on left $\times$ fingers UP on right: $3 \times 2 = 6$. That’s the UNITS. → **6.**”
 - “Total: $50 + 6 = 56$. ✓”

- Repeat for 6×9 :
 - 6: 1 down, 4 up. 9: 4 down, 1 up.
 - Down sum: $1 + 4 = 5 \rightarrow 50$.
 - Up product: $4 \times 1 = 4$.
 - Answer: **54**. ✓
- Repeat for 8×9 :
 - 8: 3 down, 2 up. 9: 4 down, 1 up.
 - Down sum: $3 + 4 = 7 \rightarrow 70$.
 - Up product: $2 \times 1 = 2$.
 - Answer: **72**. ✓

Practice (12 min)

- Whole class: teacher calls 7×6 . Children put up hands and find the answer (42).
- 5 problems chorus: 9×9 (81), 6×7 (42), 8×8 (64), 9×6 (54), 7×9 (63).
- Pair drill (last 7 min):
 - Partner A calls a problem (any from 6×6 to 9×9).
 - Partner B answers using the finger trick.
 - Swap.

Show + chant (5 min)

- “Who got $9 \times 9 = 81$ using their hands?” — hands up.
- Final chant.

Homework

- Teach the finger trick to a parent or sibling. Multiply 7×8 together.

Differentiation

- **Tier up:** Try 5×6 — what does the trick give? (5: 0 down, 5 up. 6: 1 down, 4 up. Down sum $1 \rightarrow 10$. Up product $5 \times 4 = 20$. Total 30. ✓) Notice: works for 5+ too. What about 5×5 ? (Both 0 down. Down sum 0. Up product $5 \times 5 = 25$. ✓ Total 25.) Cool — works for 5 too.
- **Tier down:** Stay with 7×8 and 9×9 until comfortable. Don’t push variety today.

Teacher’s quick notes

- A few children will have trouble keeping track of “down vs up.” Slow down with them; the trick *is* tricky on the first day.
- Some children will object: “I already know my 7 times table.” Honest response: “*Great! Now you have TWO ways of finding the answer.*”
- Don’t explain why it works today. Day 4 is the dot-drawing day.

Sanskrit term mention

(Hold for Day 6.)

Day 4 — Why the Trick Works (Dot Drawing)

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children draw a dot pattern showing why $7 \times 8 = 56$ and see the connection between fingers and the answer.

Materials

- A4 paper per child (or larger).
- Coloured crayons / pencils (2 colours per child works fine).
- The hand poster from Day 3.

The flow

Settle + chant (3 min)

Quick Twins chant + 5 finger-trick chorus answers.

Story (7 min) — “Let’s see WHY”

- “Yesterday I told you the trick. Today I’ll show you why it works.”
 - Draw a 7×8 rectangle on the board: 7 columns, 8 rows.
 - Count the total dots together: 7 columns $\times$ 8 rows = 56 dots.
 - Now divide the rectangle into regions:
 - A 5×5 **block** in one corner (the “fully solid” block).
 - A 5×3 **strip** below (because $8 = 5 + 3$).
 - A 2×5 **strip** to the side (because $7 = 5 + 2$).
 - A 2×3 **corner** (fingers UP $\times$ fingers UP).
- Total dots: $25 + 15 + 10 + 6 = 56$. ✓
- “See the 2×3 corner? That’s the fingers UP — and $2 \times 3 = 6$, the units of our answer.”
 - “See the 5×5 block plus the two strips? Together they make 50 — the tens of our answer.”

Drawing (15 min)

- Each child draws a 7×8 dot rectangle on their A4 paper.
- Colour the 2×3 corner one colour (the “fingers up” region).

- Colour the rest a different colour (the “tens” region).
- Count and label: “fingers-up corner = ” **and** ”rest = .”

Show + chant (5 min)

- 2 children share their drawings.
- Final chant.

Homework

- Pick another problem (6×7 OR 9×8) and draw the dot rectangle. Colour the corners. Bring it tomorrow.

Differentiation

- **Tier up:** Draw the same picture for 9×9 . The corner is $1 \times 1 = 1$ dot. The rest is 80 dots. Total = 81. ✓
- **Tier down:** Pre-printed grid; child just colours the regions.

Teacher’s quick notes

- The drawing IS the proof. Children who see the picture once will trust the finger trick forever.
- Some children will try to colour outside the lines, or count dots wrong. That’s fine — the colouring is the point, the counting is the practice.
- Don’t introduce algebra. The dot drawing is the entire “why” for this band.

Day 5 — Pairs Race

Module: Multiply-Near-Ten (Primary) · **Time:** 30 min

Goal

Children speed up the finger trick to under 5 seconds per problem.

Materials

- A “Race Sheet” — 16 problems (all combinations 6×6 through 9×9) on a single page.
- Stopwatch.

The flow

Settle + chant (3 min)

Twins chant + 5 finger-trick chorus answers.

Race round 1 (8 min)

- Pass out the Race Sheet (face-down).
- On “Go,” flip and solve all 16 problems using the finger trick. As fast as possible.

- After 3 minutes, “Stop.” Note how far each child got and how many were correct.
- Self-correct against the answer key.

Race Sheet:

$6 \times 6 = \underline{\quad}$	$7 \times 6 = \underline{\quad}$	$8 \times 6 = \underline{\quad}$	$9 \times 6 = \underline{\quad}$
$6 \times 7 = \underline{\quad}$	$7 \times 7 = \underline{\quad}$	$8 \times 7 = \underline{\quad}$	$9 \times 7 = \underline{\quad}$
$6 \times 8 = \underline{\quad}$	$7 \times 8 = \underline{\quad}$	$8 \times 8 = \underline{\quad}$	$9 \times 8 = \underline{\quad}$
$6 \times 9 = \underline{\quad}$	$7 \times 9 = \underline{\quad}$	$8 \times 9 = \underline{\quad}$	$9 \times 9 = \underline{\quad}$

Answer key:

36	42	48	54
42	49	56	63
48	56	64	72
54	63	72	81

Race round 2 (7 min)

- Same sheet, fresh strip. Solve as many as possible in 2 minutes.
- Beat your round-1 count.

Pair quick-fire (8 min)

- Pair up.
- Partner A reads a problem from the sheet. Partner B answers within 5 seconds (using fingers).
- Swap. Cover 10 problems each.

Wrap-up (4 min)

- Hands up: who got all 16 right? Who beat their round-1 score?
- Cheering. Final chant.

Homework

- Practice 6×6 through 9×9 at home. 5 minutes a day. Time yourself.

Differentiation

- **Tier up:** Try problems with at least one factor being 5 (e.g. 5×8 , 5×9). The trick still works! And: do all 16 in 60 seconds.
- **Tier down:** 8 problems instead of 16. Half the sheet is fine.

Teacher’s quick notes

- Don’t make this competitive between children. “*Race yourself, not your neighbour.*”
- A child who got 4/16 should celebrate — that’s progress from yesterday’s 0/16.
- Some children will revert to memorised times-table answers. That’s fine, but ask them to confirm with the finger trick.

Day 6 — Why It Works — The Algebra

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Derive $(B-a)(B-b) = B(B-a-b) + ab$.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *vinkulam* (IAST: vinkulam; lit. “deficit (number below base)”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Why It Works — The Algebra**. Keep sentences short.

Activity (8 min)

A simplified version of *Method Speed Race* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — When To Use It

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Identify problems where Nikhilam is faster than long multiplication.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *ādhikya* (IAST: ādhikya; lit. “excess (number above base)”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **When To Use It**. Keep sentences short.

Activity (8 min)

A simplified version of *Method Speed Race* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Speed Sprints

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Race traditional vs Nikhilam on 30 problems.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

Today's idea (10 min)

- *Ūrdhva-tiryagbhyāṁ* (IAST: Ūrdhva-tiryagbhyāṁ; lit. "vertically and crosswise" — the general-purpose sūtra)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Speed Sprints**. Keep sentences short.

Activity (8 min)

A simplified version of *Method Speed Race* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Mixed Practice + Module 7 Crossover

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Solve problems combining $\times 11$ and near-base methods.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

Today's idea (10 min)

- *Nikhilam* (IAST: Nikhilam; lit. "the Nikhilam sūtra (also used in Module 9 for subtraction)")

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Mixed Practice + Module 7 Crossover**. Keep sentences short.

Activity (8 min)

A simplified version of *Method Speed Race* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Final Assessment

Module: Multiplication Near a Power of 10 — Nikhilam (Primary) · **Time:** 30 min

Goal

Summative quiz and method-comparison reflection.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *vinkulam* (IAST: vinkulam; lit. “deficit (number below base)”)

Introduce the concept gently using a story, a drawing, or a sensory example. The concept for today: **Final Assessment**. Keep sentences short.

Activity (8 min)

A simplified version of *Method Speed Race* (see activities/). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Formative Quiz — Multiplication Near a Power of 10 — Nikhilam (Primary)

When: mid-module (after Day 5). **Duration:** 20 minutes. **Format:** mix of short answer, multiple choice, and reflection. Sticker self-assessment for Primary.

Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.

2. State the central idea of Multiplication Near a Power of 10 — Nikhilam in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Two-Digit $\times$ Two-Digit Near 100 (Above)’? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Multiplication Near a Power of 10 — Nikhilam is purely a religious concept and has no scientific framing.*
5. Connect Multiplication Near a Power of 10 — Nikhilam to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘Nikhilam’ from related concepts? Give one example.
8. (Short answer) In your own words, explain Day 3’s concept.
9. Identify **one** common misconception about Multiplication Near a Power of 10 — Nikhilam and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

How this is used

This quiz is **formative** — it’s a check-in to see what’s landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

Summative Quiz — Multiplication Near a Power of 10 — Nikhilam (Primary)

When: end of module (after Day 10). **Duration:** 30 minutes. **Total points:** 20.

Questions

Part A — Vocabulary (~5 points) A1. Define *Nikhilam* in your own words. A2. Define *vinkulam* in your own words. A3. Define *ādhikya* in your own words. A4. Define *Ūrdhva-tiryagbhyām* in your own words.

Part B — Concepts (~10 points) B1. Explain the central idea of Multiplication Near a Power of 10 — Nikhilam in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Multiplication Near a Power of 10 — Nikhilam with a concept from another module (your choice). What’s similar, what’s different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

Part C — Application (~5 points) C1. Give an example of Multiplication Near a Power of 10 — Nikhilam from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

Part D — Reflection (~5 points; ungraded for content) D1. What surprised you most in this module? D2. What’s one question you still have?

Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

Pass mark

A score of 12/20 indicates the student has met the module's learning objectives.

Homework

Homework — Multiplication Near a Power of 10 — Nikhilam (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — Finger-Trick Prelude** · Draw about today's concept. (~5 min)
2. **Day 2 — Two-Digit × Two-Digit Near 100 (Below)** · Draw about today's concept. (~5 min)
3. **Day 3 — Two-Digit × Two-Digit Near 100 (Above)** · Draw about today's concept. (~5 min)
4. **Day 4 — Mixed — One Above, One Below** · Draw about today's concept. (~5 min)
5. **Day 5 — 3-Digit × 3-Digit Near 1000** · Draw about today's concept. (~5 min)
6. **Day 6 — Why It Works — The Algebra** · Draw about today's concept. (~5 min)
7. **Day 7 — When To Use It** · Draw about today's concept. (~5 min)
8. **Day 8 — Speed Sprints** · Draw about today's concept. (~5 min)
9. **Day 9 — Mixed Practice + Module 7 Crossover** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

Activity 1 — Method Speed Race

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Primary · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Multiplication Near a Power of 10 — Nikhilam through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

20 problem cards, stopwatches, two columns on whiteboard

What students do

Split class in two: one team solves with traditional long multiplication, the other with Nikhilam. Race on identical problems. Discuss which problems favour which method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Personal Mental-Math Toolkit

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Primary

What you will do

Each student compiles a one-page toolkit: when to use $\times 11$, when Nikhilam multiplication, when traditional long-mult. Include 3 worked examples for each method.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a drawing / made object).
- **Written reflection** (oral share): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (3-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 1 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

Rubric

Assessment Rubric — Multiplication Near a Power of 10 — Nikhilam (Primary)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

Criteria

- **Knows the new word(s)** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Joined in activities** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Asked good questions** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Drew / made something** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice

How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.
3. Share at least one strength and one area for growth with each student.

Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.